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智動化揀貨系統中優化訂單指派基於
訂單批次處理之研究

**Order Assignment Optimization of
Robotic Mobile Fulfillment System
Based on Order Batching**

研 究 生：邱榆婷

指導教授：周碩彥 博士

郭伯勳 博士

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Master's Thesis Recommendation Form



M10801107

系所：

工業管理系

Department/Graduate Institute

Department of Industrial Management

姓名：

邱榆婷

Name

CHIU, YU-TING

論文題目：

智動化揀貨系統中優化訂單指派基於訂單批次處理之研究

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指導教授簽章：

Advisor's Signature

周正方

共同指導教授簽章（如有）：

Co-advisor's Signature (if any)

郭伯勳

日期：

Date(yyyy/mm/dd)

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Qualification Form by Master's Degree Examination Committee



M10801107

系所：工業管理系
Department/Graduate Institute Department of Industrial Management

姓名：邱榆婷
Name CHIU, YU-TING

論文題目：智動化揀貨系統中優化訂單指派基於訂單批次處理之研究
(Thesis Title) Order Assignment Optimization of Robotic Mobile Fulfillment System Based on Order Batching

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Member's Signatures

羅士哲 郭伯勳
周石 _____

指導教授簽章：

Advisor's Signature

共同指導教授簽章（如有）：

Co-advisor's Signature (if any)

系所（學程）主任（所長）簽章：

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周石
郭伯勳
林希偉
110 / 08 / 25

ABSTRACT

Robotic Mobile Fulfillment System (RMFS) is a parts-to-picker material handling system that is well known to be implemented in e-commerce warehouses. Robots play the role of picking up movable shelves, called pods, to the workstations and returning them to the storage area. Although the efficiency of the picking process is improved by reducing ineffective picking activities such as traveling and searching, moving a robot as a means of transportation consumes too much energy. Thus, fulfilling more orders with fewer robots is a crucial point to optimize the order picking process for reducing the robot movement, which can further reduce the energy consumption of robots. Fewer robots means fewer pods are picked at the picking station.

This study focuses on order to pod assignment under consideration of the order batching concept, which deals with a group of orders at the same time, in an attempt to minimize the number of picked pods and provide a high pile-on value. Furthermore, this study proposes a two-phase assignment for order to pod assignment, which can identify the pod status before the assignment.

Rather than handle orders one by one, implementing the order batching concept to the order assignment can reduce the number of picked pods by 12.64%, and increase the pile-on value from 2.87 to 5.28. The proposed two-phase assignment can reduce the number of picked pods by 18.97%, pile-on value can reach 6.37 where the throughput efficiency is still the best.

Keywords: Robotic Mobile Fulfillment System (RMFS), Simulation, Order to Pod Assignment, Order Batching, Agent-Based Modeling

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

With the rise of the Internet age, people are gradually adjusting their consumption methods and patterns, and online shopping has become one of the most popular online activities worldwide. The global e-commerce market is expected to reach 21.8% of all retail sales worldwide by 2024 [1]. Not only the growth of the e-commerce market but also the biggest challenge to warehousing operations [2]. E-commerce involves tremendous small orders, facing uncertain and large customer demands such as rapidly changing trends and promotional activities, online retailers need to fulfill orders as soon as possible [3].

To fulfill orders quickly and efficiently, warehouses play a critical role in the supply chain. The internal operations of warehouses must be executed effectively, including orders received, storage, sorting, task assignment, picking, packaging, and shipping. Among them, order picking operation is an important and tedious task. From the perspective of labor requirement, the labor cost directly related to the order picking operation even accounts for more than 50% of the total cost [4]. It can be seen that the performance of order picking operations has an overwhelming impact on the operating costs of the warehouse.

A variety of Order Picking Systems (OPS) can be distinguished in warehouses, a classification of OPS is including five main groups [5], which shows in Figure 1.1:

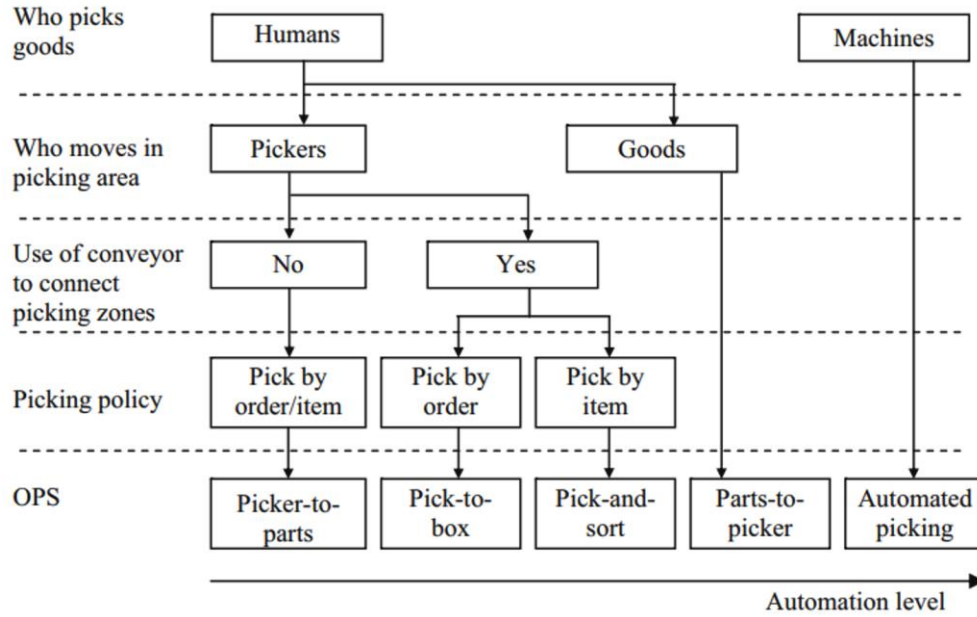


Figure 1.1 Classification of order picking systems[5]

Within the OPS, “Picker-to-parts” system is the majority of picking systems in warehouses [6]. It can be considered as the basic system for the picking activity. However, with the increasing number of customer orders in the e-commerce warehouses, the type of OPS is transformed into “Parts-to-picker” system. In order to reduce the operational cost and improving their operational productivity and efficiency, implement automation in order picking process of warehouse operations could help to reach the goal. Automated material handling systems tends to have much lower error rates than manual picking, and it improves the accuracy of inventory management and tracking, and they can achieve higher pick rates [7]. A Robotic Mobile Fulfillment System (RMFS) is an automated parts-to-picker system, where a fleet of autonomous mobile robots move between the storage area, bring the movable storage units, so-called pods, to the picking stations for reducing the handling movement of pickers to search for inventory [7, 8]. The innovation of RMFS comes from a company called Kiva Systems, which was acquired by Amazon Robotics in 2012 [9]. Compared to traditional methods, RMFS has the

flexibility necessary for handling strong demand fluctuations, which is capable to cope with large e-commerce warehouses with many SKUs (Stock Keeping Units).

The RMFS uses Automated Guided Vehicles (AGV) to bring pods, which contains ordered items, to the picking station, workers in the picking station pick items to fulfill the orders, then AGVs bring pods back to the storage area or bring them to the replenishment station to replenish SKUs on the pod. The process includes the order picking and replenishment. Throughput rate, unit of items goes through the process per unit time, is usually used to evaluate the performance of this system. It is crucial to optimize the order picking process, it means that collect more orders with less effort, for improving the throughput rate of the system. Since the performance of the system follows the determined decision rules, the unit throughput rate in this system is greatly affected by the decision rules of picking order assignment (POA) [7], which is the assignment for selecting pick orders to the picking station [10].

Fulfilling more orders with less number of robots, i.e. less number of pods to be picked is one of the crucial issues for optimizing the order picking process in RMFS. In order to effectively organize the picking operations, order batching, i.e. grouping multiple customer orders into picking orders, has been proven to be pivotal [11]. The order batching method is used for traditional manual order picking systems, picker-to-parts system, where the order picker walks or rides (e.g. on an AGV) through the aisle to pick the items in the warehouse, a set of orders are split into several subsets of order items and assigned to the pickers, for reducing the picking and traveling time.

However, in RMFS, one robot can only carry a single pod, and it is not feasible to carry all the order items in that pod at one time. Therefore, if as more as order

items are picked from one pod, it can be inferred that the number of picked pods in the picking station can be reduced, as well as the robot movement distance. The average number of order items picked from one pod, named pile-on, is the most important indicator for reducing order picking time. The higher pile-on, the fewer pods are required to be transferred between the picking station and storage area, which can reduce the number of robots required [7]. Instead of handle orders one by one, applying the order batching concept to RMFS is a more efficient way for the order assignment, which can increase the pile-on value, and reduce the number of pods be used. Therefore, the order batching concept would be applied in this research.

Meanwhile, different from previous research, the pick order assignment in this study is the assignment for selecting pods to carry the order to the picking station, which will be described as an order-to-pod assignment in the following content. This difference is based on the consideration that the first destination of the robot is the pod, and the picking station is the second destination. When assigning orders to pods, the status of pods should be considered, that is, when the assignment occurs, the candidate pods are moving on their way to the picking station or are still in the storage area and haven't been assigned to any orders. This study proposes an order-to-pod assignment in which the system can identify the pods are assigned or not before orders assign to pods, and first consolidate the orders to the assigned pods, then assign the rest of the orders in the backlog to the unassigned pods. This order-to-pod assignment process is called a two-phase assignment, which also applies the order batching concept, and the objective is to minimize the total number of picked pods.

In this study, the order-to-pod assignment applying order batching is

implemented in RMFS with different scenarios, which are batch assignment and two-phase assignment. The simulation platform is used as a tool to simulate the overall operation of RMFS in the warehouse. Overall, this study aims to optimize the order picking process through the order-to-pod assignment, with minimizing the number of picked pods, thereby increase the pile-on value.

1.2 Objective and Limitation

This study focuses on the order-to-pod assignment process in the RMFS, which aims to minimize the number of picked pods at the picking station, thereby increase the pile-on value and reduce the number of robots required and reduce energy consumption. Without order batching, the order is assigned in a first come first serve (FCFS) manner one by one, meaning that it is assigned to the pod immediately when it arrives at the system. Through the order batching assignment, the orders are grouped into a batch at a certain number of orders arrive, and then they will enter the assignment process together to make the order-to-pod decision at the same time. In the proposed two-phase assignment, parts of the orders in the backlog will be released to the pods which are already assigned to other orders in advance, and then the rest of the orders will be assigned to the unassigned pods, in an attempt to minimize the number of picked pods and reduce the robot movement, which can infer to reduce the energy consumption. The optimization objective is to minimize the number of picked pods. The proposed scenarios that applying the order batching concept are described in detail further in Chapter 3.

The limitation of this study is that the battery and charging process of robots have not been considered. The acceleration and deceleration of robots are also not considered in this study, it only has a constant speed.

1.3 Organization of the Thesis

This study is structured as follows. Chapter 1 presents the background and motivation of the study, as well as research objectives and the limitations. Chapter 2 provides literature review. Chapter 3 describes the process, formulates the order to pod assignment problem, and introduces the simulation platform and its parameters. Chapter 4 shows the results of different scenarios. Chapter 5 concludes this study.

CHAPTER 2

LITERATURE REVIEW

2.1 Robotic Mobile Fulfillment System (RMFS)

Robotic Mobile Fulfillment System (RMFS) is an automated parts-to-picker material handling system conceived to efficiently fulfill e-commerce orders. The system consists of pods, robots, picking stations and replenishment stations, which are all movable, identical elements, and orders represent the input that enter the system. The cooperation of multiple robots and the coordination of many other resources increase the complexity.

The process of RMFS starts when orders arrive at the system. Each picking station can handle several unfinished orders. If there are remaining capacity at one or multiple picking stations, the order is assigned to a picking station, otherwise, it is added to a pool of unassigned orders. The pod contains multiple items. The system will search the pods containing items that are required by orders and ask the robot to pick the pod. The robot will bring it to the picking station, and the picker picks the items from the pod and thus fulfills order lines from the unfinished orders at the picking station. After the picker is done with a pod, the pod is transported back to the empty location in the storage area. Pods in which items quantity reaches certain levels will trigger the system to assign a robot to bring that pod into the replenishment station [12]. After the replenishment process is finished, the robot will bring the pod back to the warehouse. The full process is shown in Figure 2.1, arrows indicate the movement of the robot and pod [7].

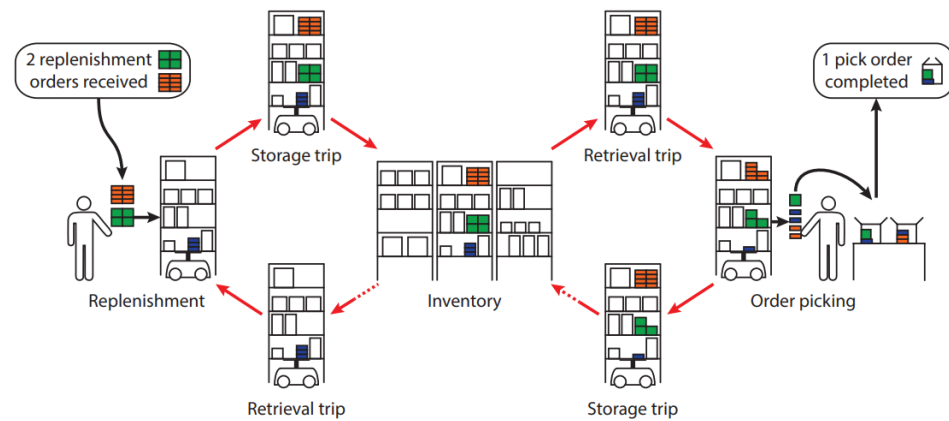


Figure 2.1 Order Picking and Replenishment Process of RMFS[7]

The decision-making issues in RMFS are interrelated. When the robot brings the pod back to the storage area, the pod can be located anywhere in the warehouse, there are several empty locations for the options. A pod typically contains multiple SKUs simultaneously. Different types of items may be in the same pod, and the same type of items may also be located in different pods or scattered throughout the warehouse [13]. The system needs to make the decisions such as the location for returning pod and the pod should be assigned when orders arrive. Those decision-making problems are classified as operational problems that need to be made in real-time (Figure 2.2).

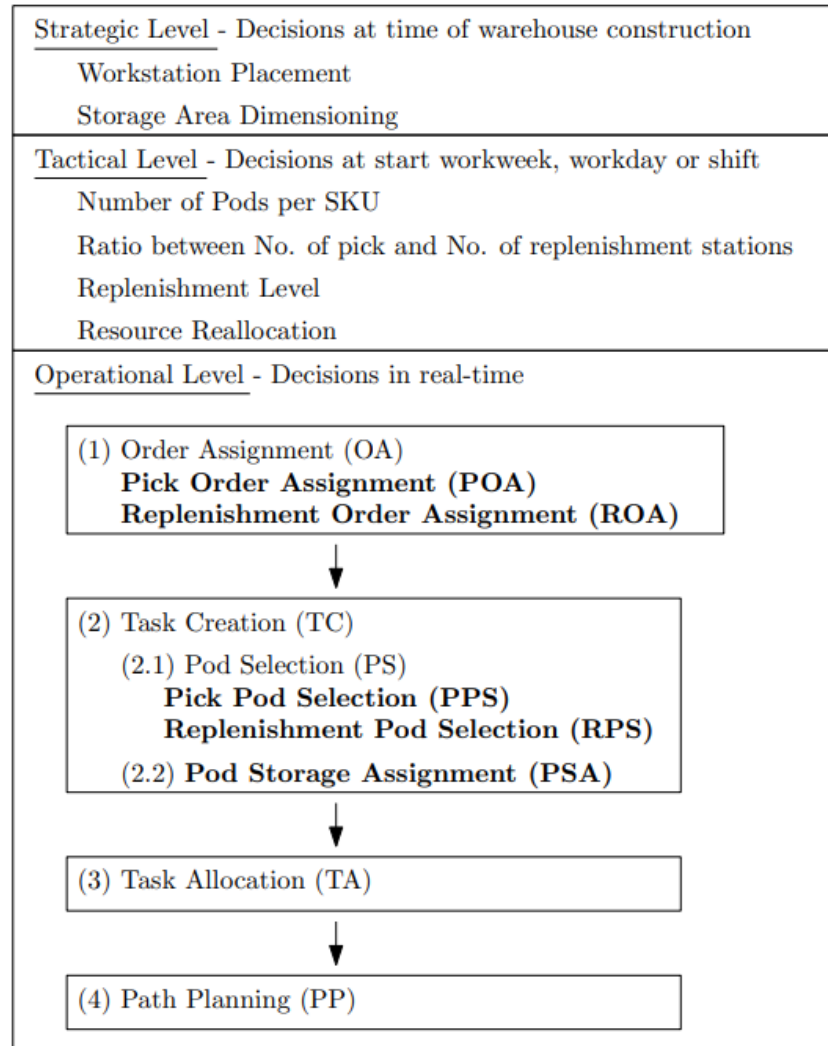


Figure 2.2 Hierarchical Overview of the Decision Problems in RMFS[7]

A research analyze optimization performance by studying important decision rules, compare multiple rules and find correlation between them to increase order item throughput [7]. In an RMFS environment, the decision problems are used in the decision-making steps, including [14]:

- Order Assignment (orders to stations)
 - Pick Order Assignment (POA): assignment of pick orders to pick stations
 - Replenishment Order Assignment (ROA): assignment of replenishment orders to replenishment stations

- Task Creation (for pods)
 - Pick Pod Selection (PPS): selection of the pods to use for picking the pick orders assigned at a pick station
 - Replenishment Pod Selection (RPS): selection of the pod to store one replenishment order
- Task Allocation (TA) (for robots): assignment of tasks from Task Creation and additional support tasks like idling to robots
- Path Planning (PP) (for robots): planning of the paths for the robots to execute

The assignment of pick orders to the picking stations (Pick Order Assignment, POA) is a critical issue for order picking optimization in an RMFS. POA is the assignment for selecting pick orders from the backlog and assigns them to the picking station [10]. Seven performance measures in RMFS are as follows: unit throughput rate, pick order throughput rate, order turnover time, distance traveled per robot, order offset, the fraction of orders that are late, pile-on, and the pick station idle time. The result shows that the POA is the most affecting decision rules for the throughput rate. It improves up to twice the throughput rate [7].

Another research studied processing the orders at the picking station. The batching and sequencing of picking orders can decrease the needs of robots. The decision problem is formulated as an NP-hard problem and solved by several heuristic algorithms to converge to the optimal. The result shows that the algorithm decreases the number of pod visits, but only for small instances. The results imply that the provided heuristic approaches would take a significant amount of time for larger instances and may not be feasible for real-time

applications [15].

Another research provides a literature review to determine the performance characteristics of RMFS. The result shows that the relationship between performance and RMFS operational design decisions is a preliminary finding of an ongoing study [16].

2.2 Order Batching

In the traditional manual picking warehouses, the order pickers depart to the storage area to retrieve/pick items after receiving orders (picker-to-parts system). When the order is large, a single order picking strategy may be applied, i.e., one order is picked at a time. However, when the order is small, this kind of picking strategy may cause the total order picking time to become longer. Therefore, picking a group of orders in a single picking journey can reduce travel time [6]. Order batching is the method that grouping a set of customer orders into picking order. This kind of method can be classified as an NP-hard problem, which can improve warehouse efficiency by reducing operating costs [17].

The objective of order batching is to minimize the traveling time and distance in the manual picking process. In terms of the availability of customer order information, order batching can be distinguished into static batching and dynamic batching [18]. In the static case, the characteristics of each customer order, i.e., the requested items and the corresponding quantities, are known when the batching decision is taken. The issue for order batching is to decide which orders should be assigned to a batch. With regard to dynamic batching, order arrives stochastically and the information of order becomes available only when it has arrived, which means customer orders arrive at different points in time while the picking process

is already being executed. The problem of dynamic batching is to determine the batch size or batch time window to minimize the expected throughput time of the order.

There are basically two criteria for batching: the proximity of pick locations and time windows [19]. Proximity batching means assigning orders to a batch based on the proximity of their storage locations. The major issue for proximity batching is how to estimate the proximities among orders, it implicitly assumes the rule for picking sequencing to visit a set of storage locations. There are some researches that studied order batching problem based on proximity batching, they all aim to minimize the processing time [20], including through minimizing the total traveling time [21], traveling distance [22] [23] [24], and also the maximum lead time [25]. Another study focused on maximizing the total association of order batches [26], considering the order batching as a NP-hard problem, and solved it by heuristic methods, such as branch-and-bound algorithm [25], genetic algorithm [22], ant colony optimization [23] and tabu search [24]. With respect to time window batching, the orders arriving during the same time interval (fixed or variable length), called a time window, are gathered as a batch. These orders are then processed simultaneously in the following stages. Some of the studies consider variable time window order batching with stochastic order arrivals for manual picking systems, which means that the number of items per batch is fixed [27] [28] [29] [30], modeling the problem as a batch service. For each possible picking batch size, they first estimate the first and second moments of service time. Then using these moments, they can find the time in a random order system. Then determine the best pick batch size in a straightforward manner. The results of simulation experiments show that this method provides high accuracy. Furthermore, it is simple and can be

easily applied in practice [6].

As mentioned above, most existing publications aim to minimize the order picker traveling time by batching the orders. However, a large batch size results in a shorter average processing time for each customer order, but it also leads to a large average waiting time. On the other hand, the average order processing time is large for small batch size, whereas the average waiting time is shorter. There is a trade-off between order batch size and waiting time (throughput time) [28]. In the manual order picking system, the order batch is assigned to the picking station, and the batch size is determined based on the required time to pick the whole batch completely [30]. The objective of minimizing the order picker traveling time alone has limitations and may not represent the performance of order picking [31].

In terms of order processing time, it needs to take into account the order due time, and the tardiness of customer orders should be involved with order batching instead of using the total processing time as a measure of the quality of the solution [32]. In order to avoid production delays, the accuracy of the delivery time is an essential performance indicator for warehouses [33]. When batching customer orders and routing pickers, compliance with order due dates is a critical issue [34] [35]. Some studies consider the order batching problem in a man-on-board system with minimizing the penalties and tardiness as respective objectives, propose a heuristic that first establishes batches and then determines the release times for the batches [32] [36]. Another research focuses on customer response times by considering batching and picking operations jointly. The research observes a tradeoff between the necessary travel time and the length of the time period for which the customer order stays in the system, propose a two-step heuristic to solve the batching and routing problem simultaneously [37]. Most studies show that the

positive difference between the order due time and the batch completion time of the assigned order [34] [38]. Reducing the processing time and travel time for constructing a batch is an important fact to reduce costs and minimize customer response delays. On the other hand, ignoring the due date of the order in the process of order batching will cause a huge dissatisfaction in customer expectations regarding the efficient responsiveness of the company. It needs to be emphasized that the order due date should be regarded as an important factor in order batching [39].

Besides manual picking systems, there are some studies that discuss the order batching problem based on automated material handling systems, with integrating other factors in the smart warehouses. The storage assignment and order batching problem in the RMFS are studied by [40]. The storage assignment model aims to decide which product to put in which pod to maximize the product similarity and the order batching model aims to minimize the number of visits of pods. This is the first article that considers the number of visits of pods as an objective function. For solving the order batching problem, the research proposed a heuristic, where a batch schedule is initialized with the objective of maximizing the order association or minimizing order alienation and improved by variable neighborhood search. The result shows that the optimized storage assignment and optimized order batching can noticeably reduce the number of visits of pods.

Since the order batching problem is regarded as one of the decision-making problems, it may be an option to integrate with other order picking planning problems and solve it. A research provides a decision support tool that integrates and solves the three main operational order picking planning problems, i.e., order batching, picker routing, and picker scheduling. The objective is to increase order

picking efficiency while ensuring a high customer service level. The customer order due times, limited availability of order pickers, and high-level storage locations are considered in the planning problem, to ensure the applicability in practice. An iterated local search algorithm is introduced to solve the problem effectively and efficiently. The result shows that solving the integration problem can reduce the order picking time of a real-life case by 17% [41].

In order to reduce the unproductive picker working time as in traditional picker-to-parts warehousing systems, the research creates an AGV-assisted mixed-shelves picking systems, which combined the mixed-shelves storage policy (i.e., items of the same SKU are spread over several shelves in a warehouse) with AGVs, which brings items from the storage area to workstations instead of human pickers. The research integrates the order batching and routing problem and solves it by the variable neighborhood search algorithm. In this system, the pod (shelf) is assumed to correspond to a customer and each batch corresponds to a vehicle, each customer will be assigned to exactly one vehicle. For creating order batch, the system will select order by calculating the minimum possible distance between random orders. The result shows that the mixed-shelves storage policy is more suitable than the usual storage policy in AGV-assisted mixed-shelves systems (saving up to 67% on driving distances for AGVs) [42].

Another study proposed a task planning approach for RMFS based smart warehouses, with the aim of utilizing system resources efficiently and adaptively. Developing a task conversion algorithm that generates tasks from a batch of orders and provides a high pile-on value, and heuristic model for assignment of tasks to robots. The research considers various system dynamics such as location and battery level of robots, replenishment of new stock in various pods, utilization level

of totes, and time elapsed after generation of order tasks. For solving the problems, it develops a highly realistic simulation environment that can be considered as a digital twin of the real system and takes into account all realistic considerations such as routing, queuing, turning, energy consumption, and charging of the robots. To obtain insight into how to design a smart warehouse system with high efficiency, the research also analyzes the optimal value of order batch size and the effect of important system parameters such as robot count, order count, and SKU. The results show that the reduction in total completion time of orders is about 40% in a warehouse with 2 pick stations and 40 totes, and about 46% in a warehouse with 3 pick stations and 60 totes [43].

For a comprehensive overview of the order batching problem, there are several studies that give a literature review for order batching operations in the framework of order picking process, examine several research papers and existing reviews about order batching and order picking [44], and provide a discussion about order batching, sequencing, and picker routing problems in order to identify research trends and gaps to meet real conditions of warehouses operations [45]. With regard to different solution approaches, other researches analyze and summarize the range and idea of order batching by giving a review of the different solution approaches that have been suggested in the literature [46], and provide a detailed state-of-the-art overview of the different solution methods that have been published in the literature [18].

CHAPTER 3

METHODOLOGY

In this chapter, the process flow of the overall system is described in the first section. The order to pod assignment problem and its scenarios is illustrated in the second section. Moreover, the system configuration of the simulation is shown in the third part. The performance analysis is also included in the last part of this chapter.

3.1 Process Flow

The RMFS process flow used in this study is shown in Figure 3.1. The overall process consists of the events between 6 elements, including order, SKU, pod, robot, picking station, and replenishment station.

The picking process starts from an order arrives at the system, until the robot brings the pod to the picking station to finish the order; in the replenishment process, the pod may need to go to replenish when the inventory is below a certain level and be brought back to the storage area. The detailed explanation of each event is described as follows:

- Order arrival

As the main input of the system orders continuously enter the system following a specific distribution of their inter-arrival time.

- SKU to Pod assignment

For the inventory strategy, applying the ABC rule to classify SKUs in the warehouse. The proportion of the number of units is 60% of class A, 25% of class B, and 15% of class C. Different types of SKUs are mixed in each pod.

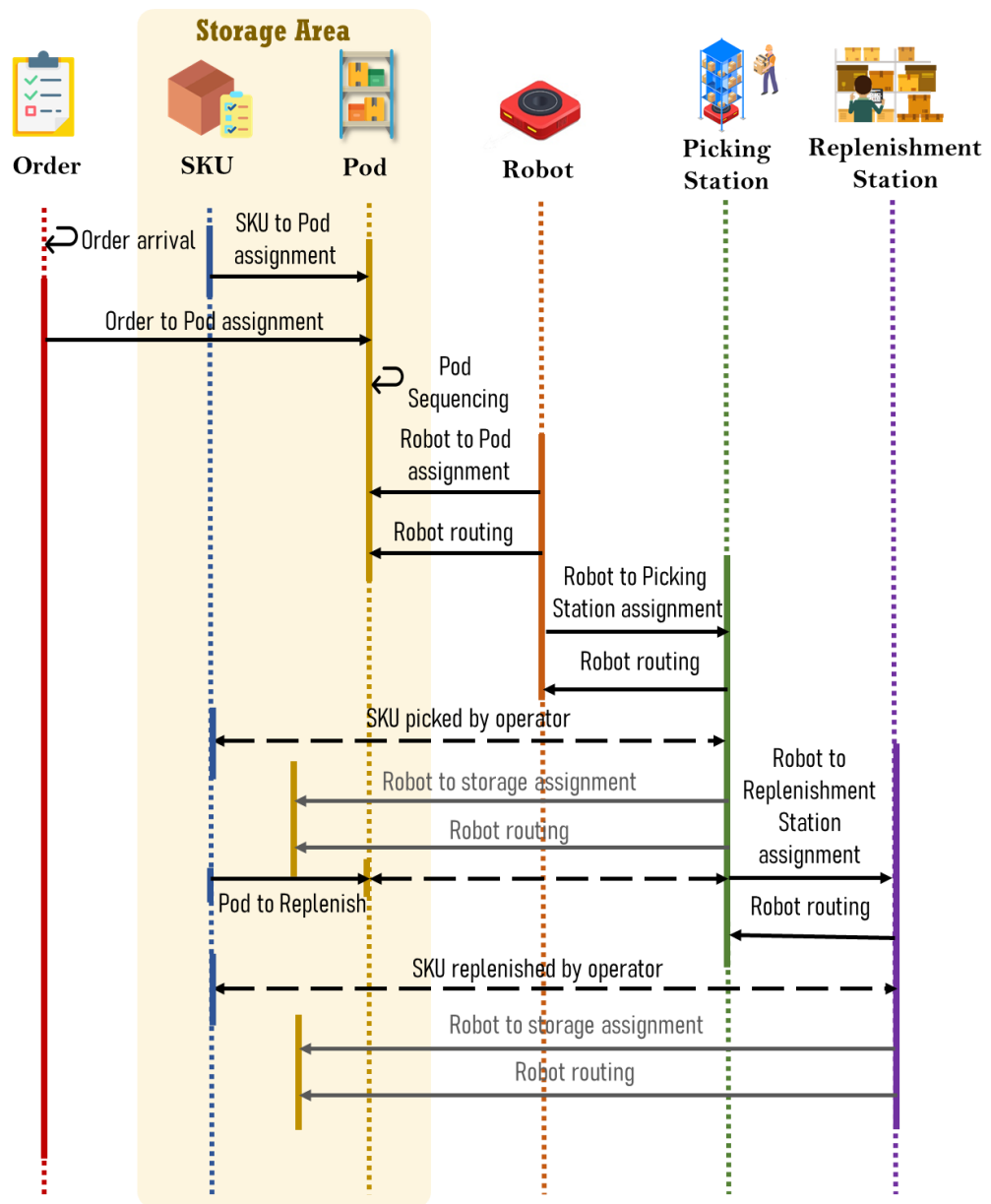


Figure 3.1 RMFS process flow

- Order to Pod assignment

After orders arrive in the system, each order will be assigned to the pod, each pod can contain multiple orders. In this study, there are three different scenarios to conduct the order to pod assignment, including first come first served (FCFS), order batching assignment, and two-phase assignment. The detailed scenario description will be explained in

Section 3.2.1.

- Pod sequencing

After the pod is assigned to the order, a list of selected pods will be sorted according to the earliest due date of the order. The sequence of the selected pod indicates the pod priorities to be picked.

- Robot to Pod assignment

AGVs will be assigned to the selected pods based on the nearest distance and earliest due date. Take twice the number of available AGVs from the list of selected pods and the assignment is conducted between AGVs and the selected pods. However, the distance between AGV and the selected pod is calculated by Manhattan distance, and applying the Hungarian algorithm to obtain the assignment results

- Robot routing

The routing policy of AGV applies Simple Routing with traffic control policy, which follows the previous study [47]. The traffic policy used is deadlock and collision prevention. This simple rule in AGV routing enables it to have a minimum path without doing path planning, which can reduce the computational time for calculation.

- Robot to workstation assignment (picking/ replenishment station)

When the robot reaches the highway in the storage area before going to the workstation, the assignment between robot and workstation is based on the minimum number of AGVs that queuing to be processed.

- Robot to storage assignment

After finishing the picking or replenishment, the robot needs to bring the pod back to the storage area. The system will comprehensively consider

all the AGVs that need to go to the empty locations in the storage area and all the empty positions together, then apply the Hungarian algorithm to determine the nearest empty location from the workstation.

- **Pod to Replenish**

After the pod finishes the order picking process, it will check the pod inventory level. If the inventory level lower than the certain percentage, the pod needs to be replenished, AGV will directly bring pod to the replenishment station.

3.2 Order-to-Pod Assignment Problem Formulation

Instead of assigning the order to the picking stations, each order will be assigned to a pod. Different from the previous related work, orders would be assigned to the picking station first, and then the required pod will be called to fulfill the order. However, it seems to be inefficient and will cause some restrictions if a picking station can only open up to three or six bins to place the ordered items, and when the orders need to be assigned to the picking stations, it will have the limitation of the number of bins for the order assignment. Instead of doing that, considering the flexibility of system resources [43], assuming the picking station can handle any order, the assignment between orders to the picking station is omitted, and the direct assignment for orders to pods can increase the possibility to optimize the order picking process. In this section, the scenarios of order-to-pod assignment and the formulation of optimization problem are illustrated in detail.

3.2.1 Scenario

There are three scenarios for order-to-pod assignment: first come first served (FCFS) assignment, order batching assignment, and two-phase assignment. The detailed descriptions are as follows.

First of all, FCFS assignment is a baseline for comparing with the other two scenarios, which is an assignment that handles order one by order. The flow chart of the FCFS assignment is shown in Figure 3.2.

After the order arrives, it will be assigned to the pod immediately. The FCFS assignment is based on the earliest due date rule, which means that the order will be assigned to the pod with the earliest due date. When it comes to multiple same options, it will be assigned to the pod randomly.

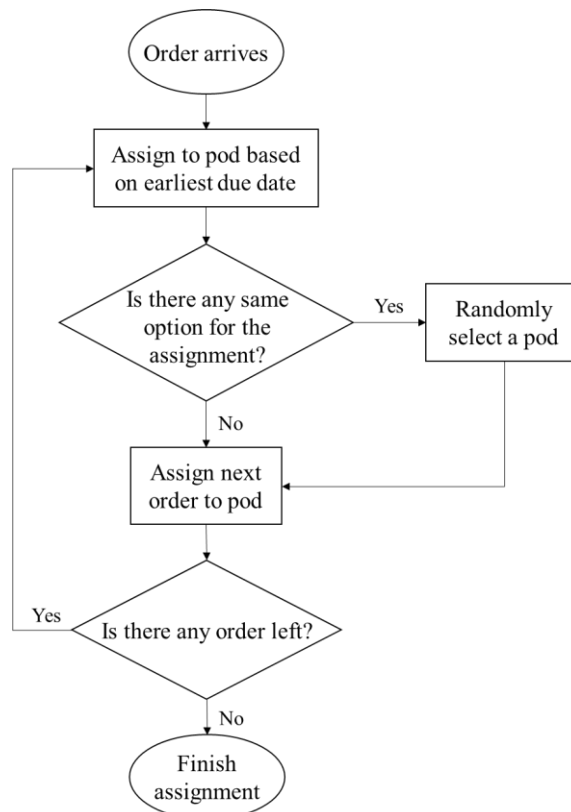


Figure 3.2 FCFS assignment flow chart

The second scenario is the order batching assignment, this scenario processes a number of orders in every assignment instead of handling orders one by one. The number of orders to be handled at once has a crucial role to increase pile-one value and in an attempt to reduce the number of picked pods as well. The flow chart of this scenario is shown in Figure 3.3.

The customer orders will arrive continuously, they will stay in the system and wait to be processed. After a certain number of orders arrive, they will be sorted by their due dates. The batch size will be determined, and the corresponded order batches will be created in order. Each order batch as a group of orders will be assigned at once in an assignment. The system will search for the pods which contain the same items as the orders in batches and its quantity is enough for the order, which means the qualified pods. The orders will be assigned to the qualified pods, and the optimization of the assignment is to assign them to pods with the minimum number. The formulation of optimization is described in the next section.

As for the batch size, since a large batch size will result in a lower throughput [28], it won't have too many orders in one batch. The next step of order to pod assignment is the robot to pod assignment, and the number of pods that could be assigned would be restricted by the number of the available robots, so if the number of orders in one batch is fixed in the first place, it could avoid the pod may not be allowed to be assigned to robots.

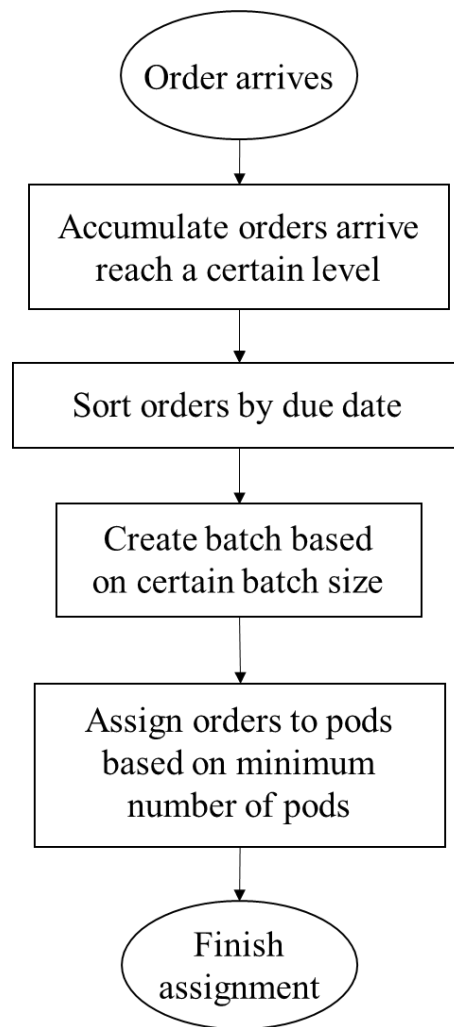


Figure 3.3 Order batching assignment flow chart

The third scenario is the proposed two-phase assignment. Same as the order batching assignment, this assignment handles a group of orders at once. The main difference between order batching assignment and two-phase assignment is the system will identify the pod status and release parts of orders to the pods which have already been assigned to other orders before doing the order batching. The flow chart of the two-phase assignment is shown in Figure 3.4.

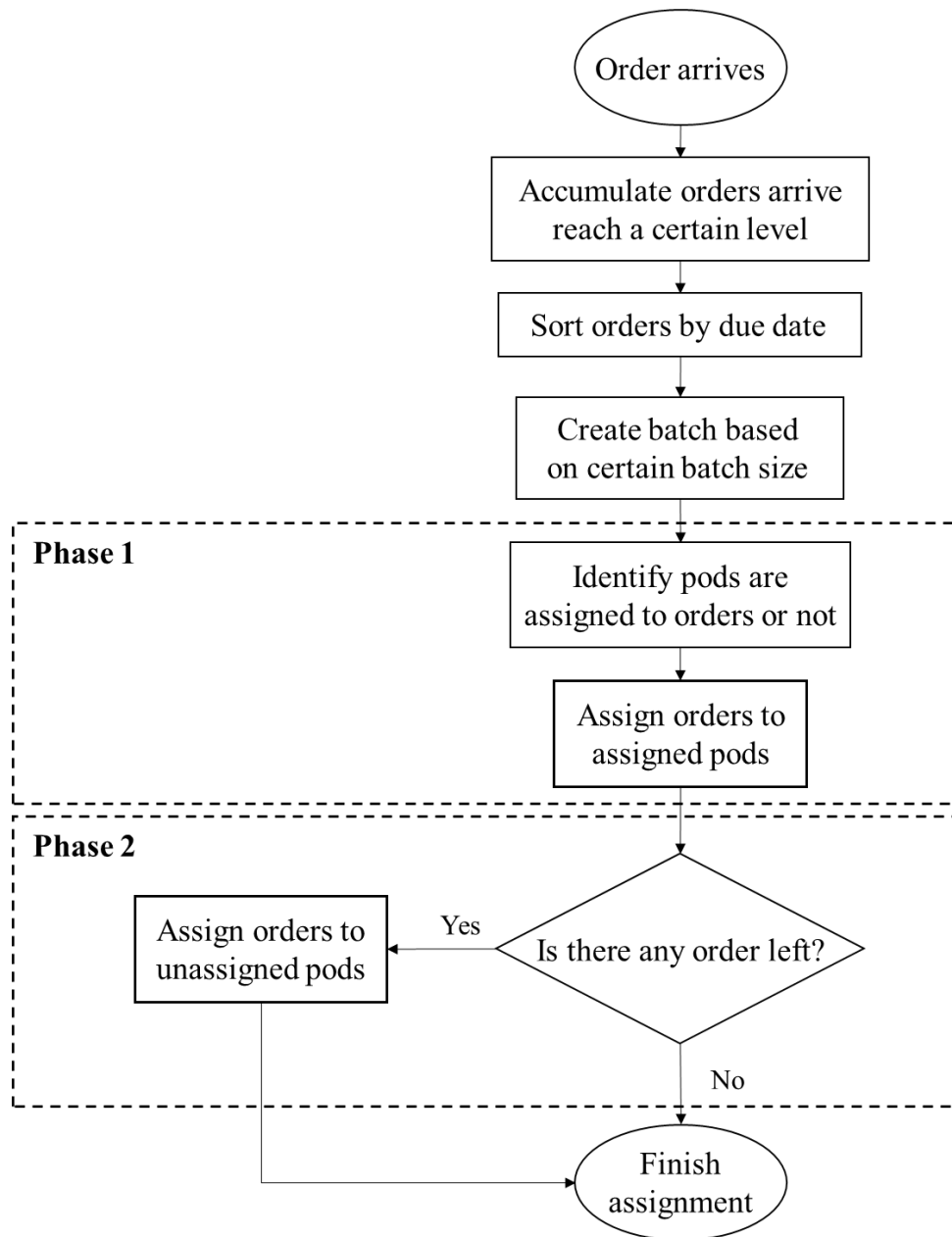


Figure 3.4 Two-phase assignment flow chart

In order to identify the status of pods, the system will differentiate the availability of pods in different steps. The status of pods is classified based on it is available to be assigned or not, and it could be divided into 6 different statuses, including:

- Assigned pod
 - Static pod: waiting to be picked by robot

- En-route pod: from storage area to the picking station
- Queueing pod: waiting in the queue
- Picking pod: picked by picker at the picking station
- Unassigned pod
 - Idle pod: no assignment yet
 - Returning pod: after being picked, on the way back to the storage area

The pod will have different statuses in the moving processes. When the pod is in the storage area and hasn't been assigned to any order, at this point it is in an "idle pod" status. After it is assigned to orders, the status will be turned into "assigned pod" and wait to be picked by a robot, so it is classified as "static pod". When the robot arrives and picks the pod, it will be brought to the picking station from the storage area, and the status will be "en-route pod". After the pod reaches the picking station, it may need to wait for other pods to be processed in the queue, so the status is "queueing pod". When the pod is picked by the picker in the picking station, the status will change into "picking pod" and not allow to be assigned anymore at this point. After the pod is done in the picking station, the robot will bring it back to the empty location of the storage area, and the status will be "returning pod". And the pod will be idle again after it is put back in the storage area. The cycle of the pod status is shown in Figure 3.5.

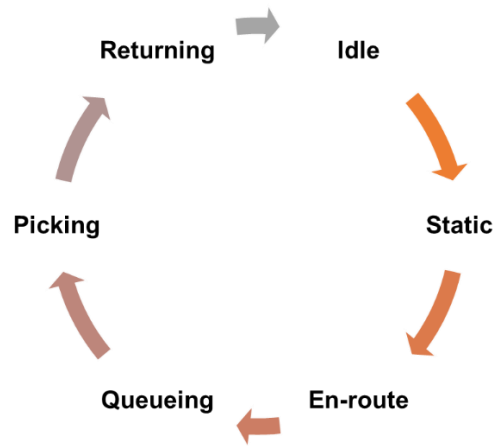


Figure 3.5 The cycle of the pod status

As for the overall process of the two-phase assignment scenario, after the order arrival and sorting. As shown in Figure 3.4, in Phase 1, the system will first identify the pod status and search for the assigned pods. If there are some orders that match the assigned pods, they will be assigned to those pods. In Phase 2, after releasing orders to the assigned pods, if there are some orders left, they will only be assigned to the unassigned pods, which are those still idle in the storage area or on the way for returning to the empty location.

3.2.2 Mathematical Model

The Order-to-Pod assignment optimization occurs when a group of orders (order batch) are assigned to pods. The notations are shown as follows:

Indices

i Order, $i = 1, \dots, m$

j Pod, $j = 1, \dots, n$

s SKU, $s = 1, \dots, k$

Parameters

q_{sj} Number of quantity for SKU s in pod j

u_{si} Number of quantity for SKU s in order i

a_{is} = 1 if order i contains SKU s ; 0 otherwise

b_{js} = 1 if pod j contains SKU s ; 0 otherwise

Decision Variables

Z_j = 1 if pod j is assigned at least one order, 0 otherwise

x_{ij} = 1 if order i is assigned to pod j , 0 otherwise

The Order-to-Pod assignment based on order batching problem can be formulated as a mixed integer programming model. The objective function (Equation (1)) is to minimize the number of picked pods.

$$\min \sum_{j=1}^n Z_j \quad (1)$$

$$\sum_{j=1}^n b_{js} Z_j \geq a_{is}, \forall i, s \quad (2)$$

$$\sum_{j=1}^n x_{ij} = 1, \forall i \quad (3)$$

$$\sum_{j=1}^n b_{js} x_{ij} = a_{is}, \forall i, s \quad (4)$$

$$\sum_{i=1}^m \sum_{s=1}^k u_{is} x_{ij} \leq q_{js}, \forall i, j \quad (5)$$

$$Z_j, x_{ij}, a_{is}, b_{js} \in \{0, 1\} \quad \forall i, j, s \quad (6)$$

The constraints are shown in Equation (2) – (6). Constraint (2) indicates that all the orders could be covered by the assigned pod. Constraint (3) ensures that each order need to be assigned to one pod. Constraint (4) states that the order should be assigned to the pod containing that SKU in the order. Constraint (5) restricts the

quantity of SKU for the set of orders which assigned to the pod could not exceed the quantity of SKU in pod. Constraint (6) is a basic constraint that restricts $Z_j, x_{ij}, a_{is}, b_{js}$ to be binary variables.

3.3 System Configuration

This section introduces the tool that used in this study, including the system architecture, simulation layout, and parameters and assumption.

3.3.1 System Architecture

In this study, the system is implemented by three integrated platforms, including NetLogo, Python, and Microsoft Excel. The system architecture is shown in the Figure 3.6. As a simulation platform, NetLogo is used to visualize the warehouse operation process in RMFS. It is an agent-based modeling and simulation that enables agents to communicate with each other. Agents can represent real entities, such as AGVs and workstations, or virtual entities like the picking process and replenishment process managers. Python is a programming language responsible for system functions like assignment results. The system functions are built-in NetLogo, which integrate with Python to obtain feedback on optimization results. As the database, Microsoft Excel provides inputs and outputs from NetLogo and Python in form of CSV files, including layout design, orders, settings, pods and items.

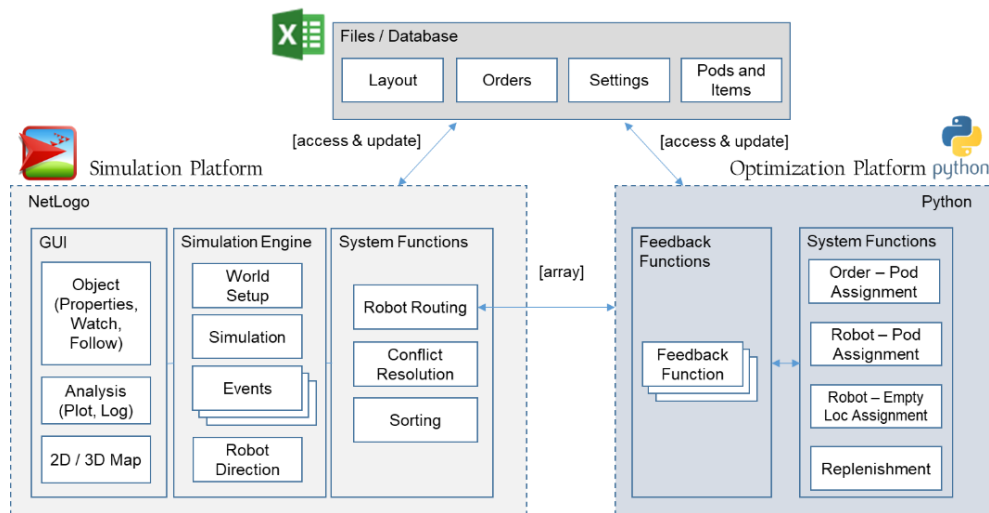


Figure 3.6 System Architecture

3.3.2 Simulation Layout

The warehouse layout consists of picking stations, storage area, and replenishment stations. Figure 3.7 shows the configuration of warehouse layout that built-in NetLogo. At the top of the layout, there are picking stations with pickers and queuing tracks. In the middle area, there is a storage area, including pods containing items. The selected pod with a different color indicates that the pod has been assigned to orders that will be picked by AGV later. The empty storage location with the blank square within pods means that AGV could bring the pod back to that position after the pod is picked. AGV can move in one-way aisle, and it is allowed to move underneath the pod without directions. In the aisle, AGV can only choose to move forward or stop. When it encounters an intersection, it can turn according to the direction of the next aisle to go. Two-lane highways are provided near the workstations to choose different directions for routes to or out of the workstations. There are two types of AGVs, which can be divided into picking AGVs and replenishing AGVs according to their functions. At the bottom of the

layout, there are replenishment stations with replenishment workers, and there are also queuing tracks.



Figure 3.7 Simulation Layout

3.3.3 Parameters and Assumptions

In order to simulate the entire process in RMFS, some information needs to be assumed. The parameters and assumptions used in this study are shown in Table 3.1.

Table 3.1 Parameters in Simulation

Parameter	Value
Simulation	
Run Length	8 hours
Replication	5 replications
Layout	
Inventory area	550 storage locations
Inventory capacity	467 pods(85% of total)
Empty storage area	83 empty storage(15% of total)
Pod batch	2 x 5 blocks
Aisles	12 vertical aisles, 4 horizontal aisles
Workstations	5 picking stations, 2 replenishment stations
Orders	
Order line	Single-line order (1 SKU, 1 quantity per order)
Initial Orders	100 orders
Orders SKU Proportion	Class A: 60%
	Class B: 30%
	Class C: 10 %
Arrival time	Exponential distribution, mean = 1.6 second
Pods	
Capacity	100 units
SKUs	
Number of SKUs	5000 SKUs
SKU distribution	Class A: 1 - 500 (10%) SKUs
	Class B: 501 – 2,000 (30%) SKUs
	Class C: 2,001 – 5,000 (60%) SKUs
SKUs/pod	20 SKUs
SKU-to-Pod distribution	Class A: 6 SKUs with 60% units
	Class B: 6 SKUs with 25% units

Class C: 8 SKUs with 15% units	
AGVs	
Number of AGVs	25 AGVs
AGV speed	1 m/s
Time to lift and store pod	1 second
Picking	
Queuing per station	5 AGVs
Picking time	Gamma distribution, $\alpha = 12$, $\beta = 1.5$
Replenishment	
Queuing per station	5 AGVs
Replenishment time	Gamma distribution, $\alpha = 19$, $\beta = 0.8$
Warehouse inventory level	60%
Pod inventory level	60%

The simulation runs for 8 hours of simulation time with 5 replications. The replication is used to reduce the impact of randomness. The warehouse layout has been shown in the previous section, and the detailed parameter settings are provided in this table. For the order data, this study uses single-line order as input, applying the ABC class rule in SKU classification. The orders are divided into three classes, and the proportion of the classes A: B: C = 6: 3: 1. The SKU-to-Pod distribution, assigning all the classes proportionally in each pod. The SKU types limit in one pod is 20 SKUs and the proportion of units is 60% of class A, 25% of class B, and 15% of class C. The cycle time in workstations follows certain distribution, and the service time is multiplied by the units be picked or replenish in the workstations. In terms of replenishment, when the warehouse inventory level is below 60% or the individual pod inventory level is below 60%, replenishment will be triggered.

3.4 Performance Analysis

In order to measure the performances, there are indicators used to compare different scenarios. The notations that are used in performance analysis are shown as follows:

Performance Notations

OR_t	Order rate in period t
TR_t	Throughput rate in period t
TE_t	Throughput efficiency in period t
P	Total number of pods
np_t	Number of picked pods in period t
P_{np}	Percentage of the number of pods used
po	Pile-on

The first indicator is pile-on value (po), which is commonly used in the performance measurement in RMFS. As mentioned earlier, pile-on means the average number of order items be picked in one pod. It is calculated by throughput rate (TR_t) and the number of picked pods (np_t) in period t . The equation is shown in Equation (7).

$$po = \frac{TR_t}{np_t} \quad (7)$$

The second indicator is the percentage of the number of pods used (P_{np}), which is the indicator that can reflect the objective of the optimization directly. The ratio of the number of pods used denotes how many pods be picked (np_t) within the total number of pods (P). The equation is shown in (Equation (8)). The lower P_{np} can

infer the reduction of robot movement and better performance.

$$P_{np} = \frac{np_t}{p} \times 100\% \quad (8)$$

The third indicator is the throughput efficiency in period t (TE_t), it is calculated by order rate (OR_t) and throughput rate (TR_t) in period t (Equation (9)). Order rate means how many orders arrive to the system in period t , and throughput rate denotes how many orders can be finished in period t . Therefore, the throughput efficiency represents how effectively the orders be finished within period t . The higher TE_t means the performance is better.

$$TE_t = \frac{TR_t}{OR_t} \quad (9)$$

CHAPTER 4

RESULT AND DISCUSSION

The experiments are conducted based on three scenarios, resulting in three indicators as the performance measurements. As mentioned before, due to the limitation of the number of robots, the batch size needs to be determined in advance. For the initial setting in order batching assignment and two-phase assignment, the batch size is set as 15, and increasing it five by five in each new test in order to find the best value of batch size.

4.1 Performance of three scenarios

First, set the batch size as 15 in the second and the third scenarios, comparing three different indicators with the baseline. Each simulation is run for eight hours, with 5 replications. The result of tests can be seen in Table 4.1.

Table 4.1 Performances of Three Scenarios

Scenario	np_t	P_{np}	TE_t	po
FCFS	348	74.52%	94.53%	2.87
Order Batching	332	71.09%	95.93%	3.45
Two-Phase	262	56.10%	96.02%	5.31

Compare to the FCFS assignment as the baseline, applying order batching assignment and two-phase assignment can reduce the number of picked pods (np_t), and the lower percentage of usage (P_{np}) is the two-phase assignment, which has only 262 pods of the 467 pods are used, nearly half of the total number of pods. For the throughput efficiency (TE_t), order batching assignment and two-phase assignment also have better performance than FCFS assignment. In previous related studies, the average pile-on value is around 1~3 [7], which means the average

number of items be picked from one pod is among 1 to 3 items. In this study, the baseline scenario also shows that the pile-on value is 2.87. With the optimization for the order-to-pod assignment, the pile-on value can be increased. In order batching assignment, it can increase more than 3, and the proposed two-phase assignment can increase the pile-on value up to 5.31.

The batch size is one of the factors that influence the throughput efficiency [28]. In order to obtain a better batch size value, this study tests different batch sizes to examine what batch size is better for the assignment. In the initial setting, the batch size is set as 15 as the previous experiment, the following are the results of the experiments for order batching assignment and two-phase assignment with batch sizes 15, 20, 25, 30 respectively. The result of the tests is shown as Table 4.2.

Table 4.2 Performances of Order Batching and Two-phase assignment in different batch sizes

Scenario	Batch size	np_t	P_{np}	TE_t	po
Order Batching	15	332	71.09%	95.93%	3.45
	20	289	61.88%	94.05%	5.28
	25	236	50.54%	86.31%	7.97
	30	196	41.97%	80.90%	9.3
Two-Phase	15	262	56.10%	96.02%	5.31
	20	259	55.55%	98.11%	6.37
	25	225	48.18%	86.76%	8.41
	30	214	45.82%	83.67%	10.28

Considering the three indicators comprehensively, the result shows that when batch size is 20, both order batching assignment and two-phase assignment have better performance. In the order batching assignment, although np_t and P_{np} have lower values and higher pile-on value when batch size is 25 and 30, their TE_t are

not good enough. In the two-phase assignment, the trend of the three indicators is similar with order batching assignment, but the throughput efficiency is higher in all the batch sizes, and the percentage of the number of picked pods and pile-on values also perform better. Both scenarios indicate that the larger batch size, the fewer number of picked pods and the higher pile-on value. When batch size is 20, the throughput efficiency of two-phase assignment can reach 98.11%, which is the best overall performance.

Since the limitation of the number of robots be assigned to pods in the following step, the batch size cannot be larger than the available number of robots. The total number of robots is 25, and as the result shows that, when batch size is 30, the performance is worse than others.

After obtaining the best batch size value in the second and the third scenario, a comparison between the three scenarios is needed. The performance of the three scenarios can be seen in Table 4.3.

Table 4.3 Performances of Three Scenarios with best batch size

Scenario	np_t	P_{np}	TE_t	po
FCFS	348	74.52%	94.53%	2.87
Order Batching	289	61.88%	94.05%	5.28
Two-Phase	259	55.55%	98.11%	6.37

Table 4.3 shows the comparison of FCFS scenarios and both order batching and two-phase scenarios with a batch size is 20. The results show that both order batching and two-phase assignment have fewer usage of pods, and also higher pile-on value than the baseline FCFS scenario. However, compared to the order batching assignment, the two-phase assignment performs better throughput and higher pile-

on value with higher batch size. With the two-phase assignment, the pile-on value can be up to 6.37, the percentage of the number of picked pods is only 55.55%, which decreases 18.97% than the baseline, and still have the best throughput efficiency at the same time.

4.2 Statistical test for the results

A statistical test is conducted based on pile-on value to prove the baseline result, and both the order batching and two-phase assignment have significant differences. First, the ANOVA test result has rejected the null hypothesis with the p-value is less than 0.05 (see Figure 4.1) and concludes that not all of the population means are equals.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
as.character(test\$group)	2	169.21	84.60	145.2	<2e-16 ***
Residuals	105	61.17	0.58		

Signif. codes:	0 '***'	0.001 '**'	0.01 '*'	0.05 '.'	0.1 ' ' 1

Figure 4.1 ANOVA Test for three scenarios

Then, using the Tukey test for the multiple comparisons, Figure 4.2 shows the Tukey test for three scenarios and Figure 4.3 shows the mean difference for each pair of groups. S1, S2, and S3 denote FCFS, order batching and two-phase scenarios respectively. From the test that is conducted, the p-values adjusted for multiple comparisons in each group are less than 0.05, which means that there is a significant difference between comparisons.

```

Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = test$pile_on ~ as.character(test$group))

$`as.character(test$group)`
      diff      lwr      upr p adj
S2-S1 1.742115 1.2901400 2.194090 0e+00
S3-S1 2.876710 2.4734271 3.279992 0e+00
S3-S2 1.134595 0.6848462 1.584343 1e-07

```

Figure 4.2 Statistic test for three scenarios

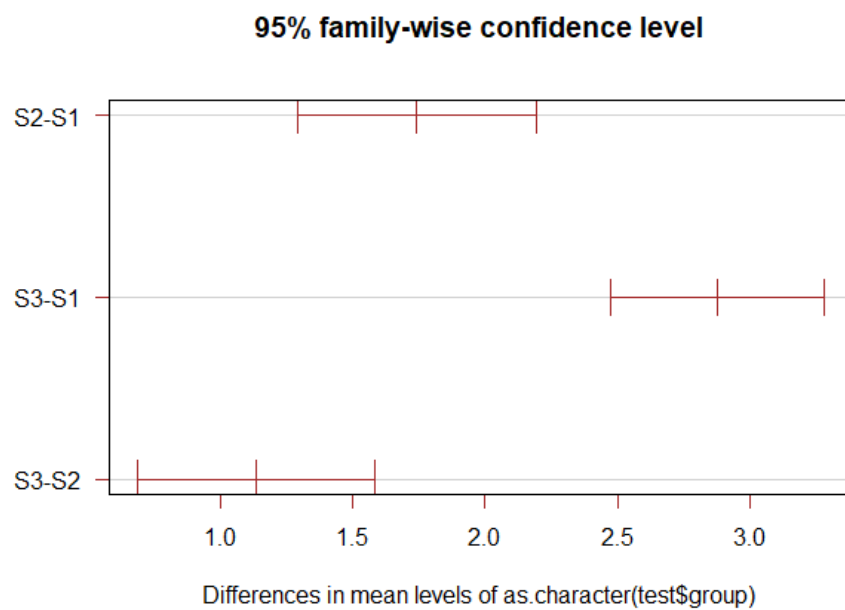


Figure 4.3 Mean difference for each pair of scenarios

CHAPTER 5

CONCLUSION AND FUTURE RESEARCH

5.1 Conclusion

RMFS is a parts-to-picker system that improves the picking rate significantly compared to the traditional methods especially for large e-commerce warehouses with many SKUs. It is critical for optimizing picking order, i.e. to collect more order items with less effort. Grouping multiple customer orders has been proven to be pivotal for an efficient way in the picking operation, with fulfilling more orders with less number of robots, which infer to less number of picked pods. The previous related research is choosing a pick order for the picking station. In contrast, this study assigns orders to pods. In this study, in order to fulfill more orders with fewer pods and increase the pile-on value at the same time, the order batching method is implemented in the order-to-pod assignment, handling orders in a group at once instead of one by one. Furthermore, the proposed two-phase assignment is divided into two steps to assign orders to pods, which first assign parts of orders to the pods that have been assigned to other orders, and then assign the rest of orders to the pods that haven't been assigned to any order in the second step. The optimization objective is to minimize the number of picked pods while increasing the pile-on value as well.

This study conducts three simulations with different scenarios, including the FCFS as a baseline, order batching assignment, and the proposed two-phase assignment for order-to-pod assignment. By implementing the order batching method for order-to-pod assignment, it can significantly reduce the number of picked pods by 12.64% compared to the baseline simulation, and increase the pile-

on value to 5.28. Moreover, the proposed two-phase assignment also includes order batching that can reduce more number of picked pods, which can reduce by 18.97% compared to the baseline, indicating that it only uses nearly half of the total number of pods. The pile-on value has also been increased significantly from the baseline of 2.87 to 6.37 in the two-phase assignment, which means around 6 order items be picked from one pod if applying two-phase assignment. Furthermore, order batch size may influence the throughput rate, however, in the two-phase assignment, the throughput efficiency doesn't drop too much, and it even reaches 98.11% when the batch size is 20. Noted that the batch size could not exceed the number of available robots, for avoiding the decrease of throughput. The proposed two-phase assignment process detects the pod status before the assignment, and then obtains better performance with higher pile-on value and fewer number of picked pods, which can reduce the robot movement and energy consumption.

5.2 Future Research

In future work, the multi-line order can be considered in the order batching strategy for optimizing the order picking process. When there are multiple SKUs and different quantities of order items, develop a different method to deal with orders for adapting the complexity is necessary. For the simulation, the charging process of AGVs could be developed. The factors of AGV movement such as acceleration, deceleration, and turning time can also be invested for deeper analysis. At the strategic level of RMFS, the warehouse layout including lots of elements, such as the number of workstations and their placement, pod batch size, and other configurations, that can be adjusted and find an optimal setting.

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